Determinant connections and closed Simpson fibers

November 5, 2026

Abstract

Let X be a smooth projective complex curve and let (E,θ) be a stable graded Higgs bundle of degree zero. We give a determinant-line proof that the flat bundles whose Simpson limit is (E,θ) form a Zariski-closed subset of the de Rham moduli space. The determinant connection of Ben-Zvi–Frenkel and the global slice of Collier–Wentworth give a single-valued primitive for the connection form in the determinant frame defined by the Simpson filtration. Consequently this frame has order zero in every algebraic one-parameter degeneration. A degree calculation for Higgs lattices then shows that the filtration extends with unchanged associated graded. The argument uses neither a compactification of the de Rham moduli space nor an extension of the connection across its boundary.

1 Introduction

Fix a smooth connected projective complex curve X of genus $g \geq 2$. Let $M_{\mathrm{Hod}}(X,r) \to \mathbb{A}^1$ denote the Hodge moduli space of semistable degree-zero rank-r λ-connections. Its fibers over 0 and 1 are $M_{\mathrm{Dol}}(X,r)$ and $M_{\mathrm{dR}}(X,r)$, respectively. The action

$$z \cdot (\lambda, V, D_\lambda) = (z\lambda, V, zD_\lambda)$$

has fixed points represented by polystable graded Higgs bundles. For a stable fixed point $x = [(E,\theta)]$, set

$$W_x = \{[(V,D)] \in M_{\mathrm{dR}}(X,r) : \lim_{z\to 0}[(z,V,zD)] = x\}. \tag{1.1}$$

Simpson's question asks whether these individual fibers are closed [5, Question 7.4].

Theorem 1.1. *For every $r \geq 1$ and every stable degree-zero graded Higgs bundle (E,θ) on X, the subset W_x is Zariski closed in $M_{\mathrm{dR}}(X,r)$.*

The argument centers on a distinguished frame of a determinant line. The stable Simpson filtration identifies its determinant of cohomology with that of the fixed graded object. On the global slice of Collier–Wentworth this gives a nowhere-vanishing holomorphic frame. The determinant connection is flat along the slice, and the slice is simply connected, so the connection form in this particular frame is exact. In an algebraic degeneration the same frame is meromorphic. Exactness forces its order at the puncture to vanish. Stability then converts this numerical statement into rigidity of the entire graded Higgs lattice.

We first carry out the argument for SL_r-connections, where both the determinant connection and the global slice are available in the form needed here. The reduction from GL_r to SL_r is made on an actual one-parameter family by taking a root of its determinant connection. All arguments at a possibly reducible limiting point take place on families of bundles; an absolute connection on the singular coarse moduli space is not required.

Conventions

Write

$$E = \bigoplus_{i=1}^{s} E_i, \qquad \theta_i : E_i \longrightarrow E_{i+1} \otimes K_X, \qquad \theta|_{E_s} = 0. \tag{1.2}$$

Choose distinct points $a_1, \dots, a_{g-1} \in X$ and put $\kappa = \mathcal{O}_X(a_1 + \cdots + a_{g-1})$. Only the equality $\deg \kappa = g - 1$ will be used. For a coherent sheaf $\mathcal{B}$ on $X \times T$ with perfect proper derived pushforward, put

$$\lambda_\kappa(\mathcal{B}) = \det Rp_{T*}(\mathcal{B} \otimes p_X^* \kappa).$$

We use the convention $\det R\Gamma = \det H^0 \otimes (\det H^1)^{-1}$. For a coherent sheaf B on X, including one with torsion, Riemann–Roch gives

$$\chi(X, B \otimes \kappa) = \deg B. \tag{1.3}$$

In particular, $\chi(X, E \otimes \kappa) = 0$. Determinant multiplicativity and its compatibility with base change are used in the sense of [3].

2 The filtration frame and its connection form

Until the reduction in Section 4.2, the connection-theoretic statements concern SL_r-connections, with $r \geq 2$, and a fixed trivialization $\det E \simeq \mathcal{O}_X$.

2.1 Simpson filtrations and the two geometric inputs

By Simpson's Rees description [5, Theorem 2.5, Lemma 4.1, and Proposition 4.3], a flat bundle represents a point of W_x if and only if it admits a filtration by holomorphic subbundles

$$V = F^0 \supset F^1 \supset \cdots \supset F^s = 0, \qquad D(F^p) \subset F^{p-1} \otimes K_X \quad (1 \leq p \leq s-1), \tag{2.1}$$

whose graded Higgs bundle is (E, θ), with $\mathrm{gr}_F^{s-i} V \simeq E_i$. The filtration is unique up to a common shift of its indices. We normalize it as in (2.1).

Every such flat bundle is irreducible. Indeed, a nonzero proper flat subbundle $B \subset V$ has degree zero. Intersecting it with $F^\bullet$ gives a graded Higgs subsheaf of (E, θ) of degree zero and rank strictly between 0 and r. Its saturation is Higgs invariant and has nonnegative degree, contrary to stability.

We use the following determinant-connection result directly.

Proposition 2.1 (Ben-Zvi–Frenkel)**.** *On the stack of SL_r-connections on X, the line $\lambda_\kappa(\mathcal{V})$ carries a holomorphic connection $\nabla^{\det}$, functorial under base change and isomorphisms of families. On the irreducible locus its curvature is*

$$F_{\nabla^{\det}} = c\, \Omega_{\mathrm{dR}} \tag{2.2}$$

for a fixed nonzero constant c, where Ω_{dR} is the trace symplectic form. The construction applies to all families of SL_r-connections, regardless of stability of their underlying bundles.

For the stated line normalization, the exact sequence $0 \to V \to V \otimes \kappa \to (V \otimes \kappa)|_{a_1 + \cdots + a_{g-1}} \to 0$ and the specified SL_r determinant trivialization give, functorially,

$$\lambda_\kappa(V) \simeq \det R\Gamma(X, V) \otimes B_\kappa, \qquad B_\kappa = \bigotimes_{j=1}^{g-1} \kappa_{a_j}^{\otimes r}.$$

The factor B_κ is a fixed one-dimensional vector space and carries the trivial connection. Proposition 2.1 is thus the determinant connection of [1, Theorem 6.2 and §§6.2.1, 6.2.3], with the fixed-curve determinant normalization of [1, §4.3]. A dual convention changes only the constant in (2.2). We will not use its construction.

We also use the global slice theorem of Collier–Wentworth [2, Proposition 1.2 and Theorem 1.4]. In the SL_r setting, there are a complex manifold S_x biholomorphic to $\mathbb{C}^N$ and a holomorphic family of flat bundles on $X \times S_x$ whose classifying map is a biholomorphism

$$p_x : S_x \xrightarrow{\sim} W_x. \tag{2.3}$$

Moreover, W_x is Lagrangian for Ω_{dR}. The family on S_x carries a filtration with fixed associated graded (1.2). To spell out this last feature, the formula for the family is

$$(\beta, \varphi) \longmapsto (\bar\partial_E + \theta^{*h} + \beta, \partial_E^h + \theta + \varphi). \tag{2.4}$$

Here $\theta^{*h} + \beta$ is strictly upper triangular in the grading (1.2), and φ is upper triangular, including its diagonal. Thus the fixed smooth filtration $F^p = \bigoplus_{i \le s-p} E_i$ is holomorphic throughout the family, and its graded Higgs maps are exactly the θ_i. This is the triangularity in [2, Definition 3.7 and equation (4.2)]. Only the holomorphic, not an algebraic, parametrization (2.3) is needed.

2.2 The distinguished determinant frame

Fix a nonzero vector

$$u_E \in \lambda_\kappa(E) \simeq \bigotimes_i \lambda_\kappa(E_i). \tag{2.5}$$

A filtered family with a graded identification with (E, θ) has a determinant identification

$$\lambda_\kappa(\mathcal{V}) \simeq \bigotimes_i \lambda_\kappa(\mathrm{gr}_F^{s-i}\,\mathcal{V}) \simeq \lambda_\kappa(E) \otimes \mathcal{O}_T. \tag{2.6}$$

Let u_{fil} denote the frame corresponding to u_E.

Lemma 2.2. *The frame u_{fil} is independent of the graded identification, provided it identifies the graded Higgs bundle with (E, θ). It is preserved by isomorphisms of filtered flat bundles.*

Proof. Two graded identifications differ by a Higgs automorphism of (E, θ). Stability makes this a common scalar a. Its action on $\lambda_\kappa(E)$ is multiplication by $a^{\chi(E \otimes \kappa)} = 1$. In a holomorphic family the same argument applies with a nowhere-zero holomorphic scalar function. The second assertion follows from functoriality of determinant multiplicativity. □

Applied to (2.4), the construction gives a global nowhere-zero holomorphic frame u_x of the determinant line on S_x. Write

$$\nabla^{\det} u_x = \alpha_x u_x.$$

Proposition 2.3. *There is a single-valued holomorphic function f_x on S_x such that $\alpha_x = df_x$. If a holomorphic family of SL_r-connections over a complex manifold T has classifying map $\gamma : T \to W_x$ and is equipped with a holomorphic Simpson filtration and graded identification, then*

$$\frac{\nabla^{\det} u_{\mathrm{fil}}}{u_{\mathrm{fil}}} = d(f_x \circ p_x^{-1} \circ \gamma). \tag{2.7}$$

Proof. By (2.2) and the Lagrangian property, $d\alpha_x = c\,p_x^*\Omega_{\mathrm{dR}} = 0$. Since S_x is simply connected, integration of this closed holomorphic one-form gives a single-valued holomorphic primitive f_x.

For the comparison, pull the slice family back by $p_x^{-1} \circ \gamma$. Locally on T, this pullback is holomorphically isomorphic to the given flat-bundle family. Indeed, trivialize the two families at a base point of X, locally on T. Monodromy matrices for finitely many generators of $\pi_1(X)$ vary holomorphically on T. Their simultaneous intertwiners form the kernel of a finite holomorphic matrix. Since corresponding fibers are isomorphic and irreducible, this kernel has constant dimension one and is a holomorphic line bundle. A local nonzero section is an invertible intertwiner and hence gives a holomorphic isomorphism of the flat families. Taking a local rth root of its determinant makes the isomorphism determinant preserving.

Uniqueness of the normalized stable Simpson filtration implies that this isomorphism preserves the filtrations. On their identified graded objects it is a common scalar. Lemma 2.2 therefore identifies the two determinant frames exactly. Naturality of the connection proves (2.7). The local identities agree on overlaps, independently of the chosen bundle isomorphisms. □

2.3 Order zero in an algebraic degeneration

Proposition 2.4. *Let (T, t_0) be a smooth pointed algebraic complex curve and let $(\mathcal{V}, \mathcal{D})$ be an algebraic family of SL_r-connections on $X \times T$, defined at t_0. Suppose that the generic fiber carries a Griffiths-transverse filtration and a graded Higgs identification with (E, θ). The resulting rational section u_{fil} of $\lambda_\kappa(\mathcal{V})$ has order zero at t_0.*

Proof. After shrinking T, the generic filtered data spread over $T \setminus \{t_0\}$. Choose a small analytic disk Δ around t_0, with parameter t. Proposition 2.1 gives a holomorphic determinant connection on the entire disk, including its central point. In a nowhere-zero holomorphic frame u_0 write

$$u_{\mathrm{fil}} = t^m a(t) u_0, \qquad a \in \mathcal{O}(\Delta)^\times.$$

This expression exists because the filtered data, and hence their determinant frame, are algebraic at the generic point. If α_0 is the connection form in u_0, then on Δ^*

$$\frac{\nabla^{\det} u_{\mathrm{fil}}}{u_{\mathrm{fil}}} = \alpha_0 + m\frac{dt}{t} + \frac{da}{a}. \tag{2.8}$$

Its integral around a small circle is $2\pi\sqrt{-1}\,m$. By Proposition 2.3, the same form has a single-valued holomorphic primitive on Δ^*, so this integral vanishes. Therefore $m = 0$. □

Remark 2.5. Meromorphicity in this argument belongs to the algebraic filtration frame. No boundedness or meromorphic extension of the primitive $f_x \circ p_x^{-1} \circ \gamma$ is asserted or needed.

3 Degree and rigidity of Higgs lattices

Let R be the local ring at a point of a smooth algebraic complex curve, let t be a uniformizer, and let K be its fraction field. Write $X_R = X \times \operatorname{Spec} R$ and let (E_R, θ_R) be the constant extension of (E, θ). A *Higgs lattice* in $E_R[1/t]$ will mean a coherent subsheaf $H \subset E_R$ satisfying

$$H[1/t] = E_R[1/t], \qquad \theta_R(H) \subset H \otimes K_X.$$

Set $Q = E_R/H$ and $Q_j = t^j Q / t^{j+1} Q$. A power of t kills Q, so only finitely many Q_j are nonzero.

Lemma 3.1. *View u_E as a rational section u_H of $\lambda_\kappa(H)$ through the generic identification $H[1/t] = E_R[1/t]$. Then*

$$\operatorname{ord}_t(u_H) = -\sum_{j\geq 0} \deg Q_j. \tag{3.1}$$

Proof. The surface X_R is regular, so the sheaves in $0 \to H \to E_R \to Q \to 0$ are perfect. Their proper derived pushforwards to R are perfect as well. Put $C_Q = R\Gamma(X_R, Q \otimes \kappa)$. This is a perfect torsion complex. The order of its canonical rational determinant section is

$$\sum_q (-1)^q \operatorname{length}_R H^q(C_Q).$$

For clarity, the sign is fixed by a module $R/(t^b)$ in degree zero: the resolution $[R \xrightarrow{t^b} R]$ in degrees $-1, 0$ gives order b. Additivity and shifts give the displayed formula for any perfect torsion complex.

The determinant identity $\lambda_\kappa(E_R) = \lambda_\kappa(H) \otimes \det C_Q$ therefore gives

$$\operatorname{ord}_t(u_H) = -\sum_q (-1)^q \operatorname{length}_R H^q(C_Q).$$

Filtering Q by powers of t and using additivity yields

$$\sum_q (-1)^q \operatorname{length}_R H^q(C_Q) = \sum_j \chi(X, Q_j \otimes \kappa) = \sum_j \deg Q_j,$$

where the last equality is (1.3). □

Lemma 3.2 (Higgs lattice rigidity)**.** *If (E, θ) is stable of degree zero and $\sum_j \deg Q_j = 0$, then $H = t^m E_R$ for some integer $m \geq 0$.*

Proof. There is a surjective Higgs morphism

$$\psi_j : E \longrightarrow Q_j, \qquad \bar{v} \longmapsto [t^j v],$$

where v is any lift to E_R. It is independent of the lift. Put $K_j = \ker \psi_j$. Multiplication by t gives $K_j \subset K_{j+1}$.

We claim that $\deg Q_j \geq 0$, with equality only when $Q_j = 0$ or ψ_j is an isomorphism $E \simeq Q_j$. If $0 < \operatorname{rk} K_j < r$, its saturation is Higgs invariant, and stability gives

$$\deg K_j \leq \deg K_j^{\mathrm{sat}} < 0, \qquad \deg Q_j = -\deg K_j > 0.$$

If $\operatorname{rk} K_j = r$, then Q_j is torsion and its degree is its length; this is zero exactly when $Q_j = 0$. If $\operatorname{rk} K_j = 0$, torsion-freeness of E gives $K_j = 0$. This proves the claim.

The hypothesis makes each $\deg Q_j$ zero. Since the kernels increase and are eventually E, for some $m \geq 0$ one has

$$K_j = 0 \quad (j < m), \qquad K_j = E \quad (j \geq m).$$

Now $Q_m = 0$ implies $t^m Q = t^{m+1} Q$, and nilpotence of t on Q implies $t^m Q = 0$. Hence $t^m E_R \subset H$. For the opposite inclusion, suppose inductively that $H \subset t^j E_R$ with $j < m$. Write a local section of H as $t^j v$. Its image under ψ_j is zero, so the reduction of v lies in $K_j = 0$. Thus $H \subset t^{j+1} E_R$. Induction gives $H \subset t^m E_R$. □

4 Extension of the filtration and proof of closedness

4.1 One-parameter extension for SL_r

Proposition 4.1. *Let $(\mathcal{V}_R, \mathcal{D}_R)$ be an algebraic family of SL_r-connections on X_R. Suppose that its generic fiber has a Griffiths-transverse filtration $F_K^\bullet$ with graded Higgs bundle identified with $(E,\theta)_K$, where (E,θ) is stable. Then its special fiber admits a Simpson filtration with graded Higgs bundle (E,θ).*

Proof. Extend each filtration step by intersection inside $\mathcal{V}_R[1/t]$:

$$A^p = \mathcal{V}_R \cap F_K^p, \qquad G_i = A^{s-i}/A^{s-i+1}. \tag{4.1}$$

The A^p are coherent. For example, if $j : X_K \hookrightarrow X_R$, then A^p is the kernel of $\mathcal{V}_R \to j_*(\mathcal{V}_K/F_K^p)$; this is a quasi-coherent subsheaf of a coherent sheaf on a noetherian scheme. The quotient $\mathcal{V}_R/A^p$ is torsion-free since it embeds into $j_*(\mathcal{V}_K/F_K^p)$. The depth lemma on the regular surface X_R therefore shows that A^p is reflexive, hence locally free. Each G_i is torsion-free, since it embeds into $\mathcal{V}_R/A^{s-i+1}$; we do not yet assert its local freeness.

Griffiths transversality extends to the A^p. Indeed, $\mathcal{D}_R A^p$ is regular in $\mathcal{V}_R \otimes K_X$ and belongs generically to $F_K^{p-1} \otimes K_X$, hence belongs to $A^{p-1} \otimes K_X$. The induced maps

$$G_i \longrightarrow G_{i+1} \otimes K_X$$

are $\mathcal{O}_{X_R}$-linear. The generic graded identifications $G_{i,K} \simeq E_{i,K}$ intertwine them with the θ_i. By coherence and quasi-compactness, multiplying all these identifications by one sufficiently large common power t^a gives injections $G_i \hookrightarrow E_{i,R}$. Consequently

$$H = \bigoplus_i G_i \hookrightarrow E_R \tag{4.2}$$

is a Higgs lattice, and its embedding respects the displayed blocks.

Determinant multiplicativity along (4.1) identifies $\lambda_\kappa(\mathcal{V}_R)$ with $\lambda_\kappa(H)$. The common factor t^a does not change the generic determinant identification: its action on the total determinant is

$$t^{a\sum_i \chi(E_i\otimes\kappa)} = t^{a\chi(E\otimes\kappa)} = 1. \tag{4.3}$$

Thus the filtration frame is precisely the rational section in Lemma 3.1. Proposition 2.4 and that lemma give $\sum_j \deg Q_j = 0$. Lemma 3.2 implies $H = t^m E_R$.

Since (4.2) is block diagonal, each G_i maps isomorphically onto $t^m E_{i,R}$. Dividing every graded identification by the same t^m gives isomorphisms $G_i \simeq E_{i,R}$ compatible with all Higgs maps. In particular, the G_i are locally free. Their successive extensions show that all $\mathcal{V}_R/A^p$ are locally free. Hence $A^\bullet$ is a subbundle filtration and its restriction to the special fiber is Griffiths transverse with graded Higgs bundle (E,θ). □

4.2 Reduction of a GL_r family to SL_r

Proposition 4.2. *Proposition 4.1 holds for families of GL_r-connections.*

Proof. For $r = 1$, the generic underlying line is E_K. Separatedness of $\mathrm{Pic}^0(X)$ implies that the special underlying line is E, which is the required assertion.

Suppose $r \geq 2$ and put $L = \det E$. The generic graded identification gives $\det \mathcal{V}_K \simeq L_K$. The morphism $\operatorname{Spec} R \to \mathrm{Pic}^0(X)$ defined by $\det \mathcal{V}_R$ is therefore constant, by separatedness. Thus

$$\det \mathcal{V}_R \simeq L_R \otimes p_R^* B \simeq L_R,$$

where the line B on $\operatorname{Spec} R$ is trivial. Fix such an isomorphism. Choose $N \in \operatorname{Pic}^0(X)$ with $N^r \simeq L$, and choose a holomorphic connection ∇_N^0 on N. Transport $\det \mathcal{D}_R$ to N_R^r. The difference between this connection and $(\nabla_N^0)^{\otimes r}$ is a regular relative one-form η. Set

$$\nabla_N = \nabla_N^0 + \frac{1}{r}\eta.$$

Then $\nabla_N^{\otimes r} = \det \mathcal{D}_R$. Twisting $(\mathcal{V}_R, \mathcal{D}_R)$ by $(N_R, \nabla_N)^{-1}$ gives an SL_r-family. Its generic filtration has graded Higgs bundle

$$(E', \theta') = (E \otimes N^{-1}, \theta \otimes \mathrm{id}_{N^{-1}}).$$

The connection on N_R contributes only a filtration-preserving term and therefore does not alter the degree-minus-one graded Higgs map. The bundle (E', θ') is stable with trivial determinant. The generic graded identification can be made determinant preserving after a finite extension of K, by taking an rth root of its determinant scalar. Normalize the base curve in this extension and work at a point above its special point. Proposition 4.1 applies there. The residue field is still $\mathbb{C}$, so its conclusion, after twisting back, is the desired statement about the original special connection. □

4.3 Passage to the coarse moduli space

Proof of Theorem 1.1. Simpson's algebraic attracting strata and their limit maps show that W_x is locally closed [5, Proposition 5.1 and its proof]. Take a framed parameter scheme with universal family and a good quotient

$$q : R^{\mathrm{ss}} \longrightarrow M_{\mathrm{dR}}(X, r)$$

as in [4, Theorem 4.7]. Put $S = q^{-1}(W_x)$ and $Z = \overline{S}$. Since S is invariant, so is Z, and the good quotient sends Z to a closed set containing W_x.

Let y be a closed point of $\overline{W_x}$. Choose a closed point $z \in Z$ above y. If $z \in S$, then $y \in W_x$. Otherwise, since S is open dense in Z, a curve through z with generic point in S exists: take successive general hyperplane sections in an affine neighborhood of z in an irreducible component of Z, and choose a component not contained in $Z \setminus S$. Normalizing that curve gives a smooth pointed curve $(T, t_0) \to R^{\mathrm{ss}}$ whose generic point maps into S. After shrinking T around t_0, we may assume that $T \setminus \{t_0\}$ maps into S. Pullback of the universal family gives an actual family $(\mathcal{V}, \mathcal{D})$ on $X \times T$, defined at t_0.

We justify the choice of generic filtered data. Over $T \setminus \{t_0\}$, consider the finite-type parameter space of subbundle filtrations with the ranks and degrees prescribed by (1.2), together with graded isomorphisms to the E_i. Such a space is obtained from relative flag Quot schemes and relative Isom spaces. Griffiths transversality and compatibility with the θ_i impose algebraic equations. Every closed fiber is nonempty by the characterization of W_x, since its points are irreducible and the universal family represents their actual isomorphism classes. The image of the parameter space is constructible and contains every closed point; its complement is therefore empty. Its generic fiber is consequently a nonempty finite-type scheme over $K = \mathbb{C}(T)$. A closed point of that fiber has finite residue extension K'/K. Thus the generic filtration and graded identification exist after a finite extension. Normalize T in K' and choose a point above t_0.

Proposition 4.2 now shows that the special flat bundle has Simpson limit x. Hence $y \in W_x$. We have proved that the constructible set $\overline{W_x} \setminus W_x$ has no closed points. A nonempty constructible subset of a finite-type complex scheme has a closed point, so this set is empty. □

Remark 4.3. Stability is used twice in an essential way: it makes the action on the filtration determinant frame independent of graded identifications, and it makes every proper positive-rank Higgs quotient have positive degree. The proof does not establish the analogous assertion for an arbitrary polystable graded Higgs bundle.

References

[1] D. Ben-Zvi and E. Frenkel, *Geometric realization of the Segal–Sugawara construction*, in *Topology, Geometry and Quantum Field Theory*, U. Tillmann (ed.), London Mathematical Society Lecture Note Series **308**, Cambridge University Press, Cambridge, 2004, pp. 46–97. doi:10.1017/CBO9780511526398.006.

[2] B. Collier and R. Wentworth, *Conformal limits and the Białynicki–Birula stratification of the space of λ-connections*, Advances in Mathematics **350** (2019), 1193–1225. doi:10.1016/j.aim.2019.04.034.

[3] F. F. Knudsen and D. Mumford, *The projectivity of the moduli space of stable curves. I. Preliminaries on "det" and "Div"*, Mathematica Scandinavica **39** (1976), 19–55. doi:10.7146/math.scand.a-11642.

[4] C. T. Simpson, *Moduli of representations of the fundamental group of a smooth projective variety. I*, Publications Mathématiques de l'IHÉS **79** (1994), 47–129. doi:10.1007/BF02698887.

[5] C. T. Simpson, *Iterated destabilizing modifications for vector bundles with connection*, in *Vector Bundles and Complex Geometry*, Contemporary Mathematics **522**, American Mathematical Society, Providence, RI, 2010, pp. 183–206. doi:10.1090/conm/522/10300.